Formal Slides

A Typst Template for Academic Presentations

Author One, Author Two

Institution Name

May 2025

typst

Table of Contents

Section 1

Typography

Default Fonts

The template ships with three IBM Plex families:

IBM Plex Serif body text Regular, *italic*, **bold**, ***bold-italic***	**IBM Plex Sans** titles & headings Regular, *italic*, **bold**, ***bold-italic***	**IBM Plex Mono** code & verbatim Regular, *italic*, **bold**

Inline styling: **bold**, *italic*, colored, <u>underlined</u>, large (13 pt), small (7.5 pt).

Text Sizing and Semantic Emphasis

Headline – 18 pt bold

Subheading – 14 pt bold

Section heading – 12 pt regular

Body text – 10 pt (default)

Caption / footnote – 8 pt muted

Semantic use: *italics for emphasis*, **bold for key terms**, red for warnings, green for results. Combine for ***critical highlighted points***.

Change the global accent via `title-color`, the secondary bands via `secondary-color`, and the page fill via `bg-color`. These parameters propagate through the navigation bar, footer, section dividers, TOC numbering squares, and theorem environments.

Section 2

Lists & Enumerations

Bullet Points and Nested Lists

Unordered list (three levels):

- First top-level item
  - Nested child A
  - Nested child B
    - Deeply nested
    - Another deep item
- Second top-level item
  - Another child
- Third top-level item

Ordered enumeration (three levels):

1. Step one: initialise
   1. Sub-step 1a
   2. Sub-step 1b
2. Step two: process data
   1. Sub-step 2a
      1. Detail 2a-i
3. Step three: evaluate output
4. Step four: report results

Typst automatically styles bullet symbols and numerals at each nesting depth. Ordered and unordered lists can be freely mixed at any level.

Section 3

Figures

Inserting and Referencing Figures

Figures use `#fs-figure(caption: [...])`. The counter resets per section; reference figures by their auto-assigned number. Longer surrounding paragraphs make it easier to inspect the vertical rhythm before and after visual material.

As shown in **Figure 1**, a colored rectangle acts as a placeholder. In practice pass `image("diagram.svg")` or any Typst content as the figure body. This text intentionally spans multiple lines so the figure margins can be judged against realistic prose.

Figure 2 illustrates that captions appear italic below the figure, numbered automatically within the current section. The paragraph below the visual block should feel close enough to belong to the same slide, but not so close that the caption looks cramped.

The first placeholder represents a diagram or image inserted into the slide flow. A few lines of prose above it help show how the template separates ordinary text from framed visual content.

Diagram / Image here

Figure 1. *Placeholder – replace with* `image("diagram.svg")`*.*

Another placeholder

Figure 2. *Second figure – captions are italic, automatically numbered.*

After the second figure, this short paragraph checks the lower margin beneath a caption. It should read as a continuation of the slide narrative rather than as text accidentally attached to the figure.

Full-Width and Fractional-Width Figures (With Captions)

A figure can expand to the full width of its column when the content should dominate the layout. This is useful for diagrams, screenshots, or images that need as much horizontal room as possible.

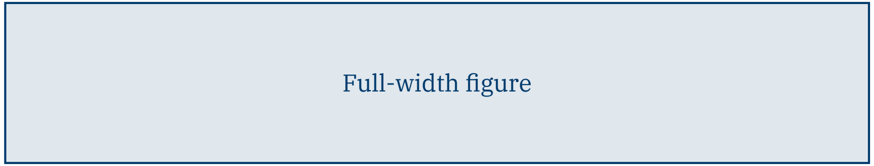


Figure 3. *Full-width placeholder figure spanning the whole column.*

The caption should stay centered under the figure even when the visual takes the entire available width of the column.

Smaller visuals often read better when they keep some white space around them. A fractional-width figure makes that possible while still preserving the same numbering and caption behavior.

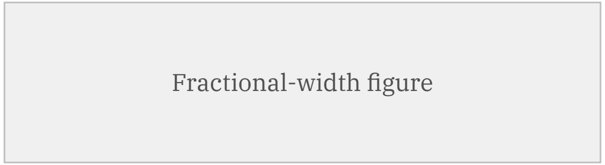
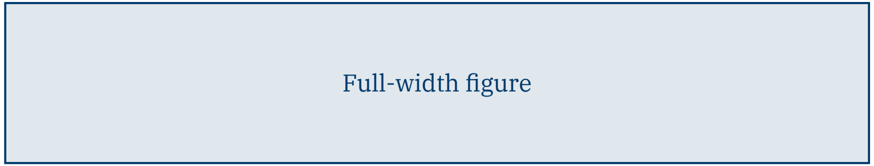


Figure 4. *Fractional-width placeholder figure centered inside the column.*

This example checks that a narrower figure still aligns cleanly in the column and that the caption feels attached to the centered image rather than to the whole column width.

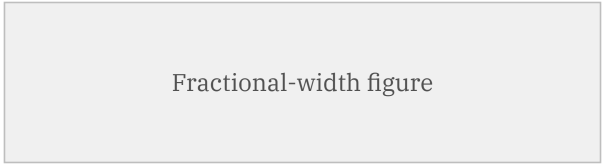
Full-Width and Fractional-Width Figures (No Captions)

The same pair can also be shown without captions when the slide is purely illustrative and the surrounding prose already provides enough context for the audience.



Without a caption, the lower margin should still separate the figure from the next paragraph and keep the slide from feeling cramped.

The fractional-width version below uses the same visual content but keeps the narrower footprint. This lets us compare centered image placement with and without the caption layer.



The surrounding text remains multi-line on purpose so we can judge the spacing around a centered narrow figure in the same way as we do for captioned figures.

Section 4

Two-Column Layouts

Two-Column Layout

Left column

The `#two-col(left, right)` helper builds a two-column `grid`. Parameters:

- `left-width` – fraction of slide width (default 48%)
- `gutter` – gap between columns (default 4%)

Works well for: text + figure, code + output, comparative tables, side-by-side theorem boxes.

Right column

Any Typst content fits inside a column, including nested `two-col` calls, theorem boxes, figures, and tables.

Arbitrary block inside a column

Section 5

Source Code & Pseudo-code

Source Code and Pseudo-code Side by Side

Source Code Python

```python
def softmax(x):
    e = np.exp(x - x.max(axis=-1, keepdims=True))
    return e / e.sum(axis=-1, keepdims=True)

def cross_entropy(logits, labels):
    probs = softmax(logits)
    n = labels.shape[0]
    log_p = np.log(probs[range(n), labels])
    return -log_p.mean()
```

Uses `IBM Plex Mono` on a light grey background. Pass a language name to `#code-box(..., lang: "...")` for syntax highlighting.

Pseudo-code Mini-Batch SGD

```
Algorithm: Mini-Batch SGD
Input: loss L, data D, lr eta, T, B
-------------------------------------
for t = 1 to T do
  B_t <- sample B examples from D
  g   <- grad_theta L(theta; B_t)
  theta <- theta - eta * g
end for
return theta
```

Pseudo-code now uses the same framed box language as the theorem environments, with a `type` label and `title`.

Section 6

Tables

Paper-style and Grid-style Tables (No Captions)

Paper style (booktabs-like – horizontal rules only)

This table is introduced by a short paragraph rather than a single label. The extra prose makes the spacing above the table visible in a realistic slide, where a table usually follows a sentence or two of setup.

Method	**Acc. (%)**	**F_1**	**AUC**
Baseline	72.3	0.701	0.743
Model A	81.5	0.803	0.851
Model B	**88.9**	**0.876**	**0.903**

Classic academic style: only top and bottom rules, no vertical lines. The text after the table deliberately runs for a couple of lines so the lower margin can be compared with the upper margin.

Grid style (full borders, colored header, alternating rows)

The grid version is meant for dense numeric summaries or dashboard-like reporting. A longer lead-in makes it easier to see whether the table feels attached to the explanation or floats too far away.

Method	**Acc. (%)**	**F_1**	**AUC**
Baseline	72.3	0.701	0.743
Model A	81.5	0.803	0.851
Model B	88.9	0.876	0.903

Dashboard style: solid grid, alternating row shading. This closing note also spans multiple lines, which helps reveal whether the table block leaves enough room before normal prose resumes.

Paper-style and Grid-style Tables (With Captions)

Paper style (booktabs-like – horizontal rules only)

Captions are useful when the table needs to be referenced later in the talk or connected to a source. This paragraph gives the captioned table enough surrounding prose to test both the top margin and the caption spacing.

Method	Acc. (%)	F_1	AUC
Baseline	72.3	0.701	0.743
Model A	81.5	0.803	0.851
Model B	**88.9**	**0.876**	**0.903**

Table 1. *Placeholder caption for the paper-style table.*

Classic academic style: only top and bottom rules, no vertical lines. With a caption present, the paragraph after the table should sit beneath the full table block rather than feeling glued to the caption.

Grid style (full borders, colored header, alternating rows)

The captioned grid table shows how a more operational table behaves inside the same layout. The text before it is intentionally longer so vertical spacing is visible without relying on empty slide area.

Method	Acc. (%)	F_1	AUC
Baseline	72.3	0.701	0.743
Model A	81.5	0.803	0.851
Model B	88.9	0.876	0.903

Table 2. *Placeholder caption for the grid-style table.*

Dashboard style: solid grid, alternating row shading. This final description should have comfortable breathing room after the caption while still reading as part of the same explanatory unit.

Section 7

Diagrams & Charts

Block Diagrams

The Transformer architecture couples an encoder stack with an autoregressive decoder. Multi-head attention and feed-forward sub-layers are wrapped in residual connections followed by layer normalisation.

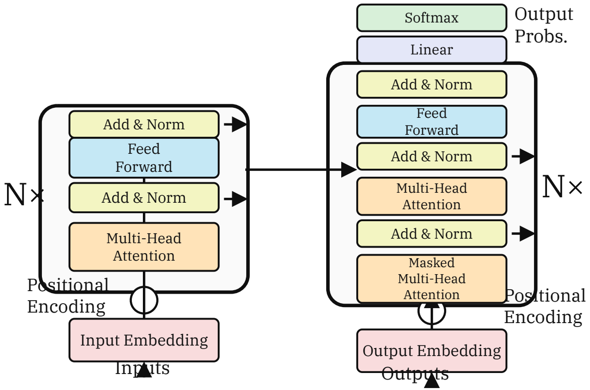


Figure 1. *Transformer encoder-decoder block diagram (Vaswani et al., 2017).*

Residual connections (Add) let gradients flow directly; Layer Norm stabilises training at each depth.

A closed-loop controller compares the reference signal with the measured output, drives the plant, and routes the response through a feedback transducer.

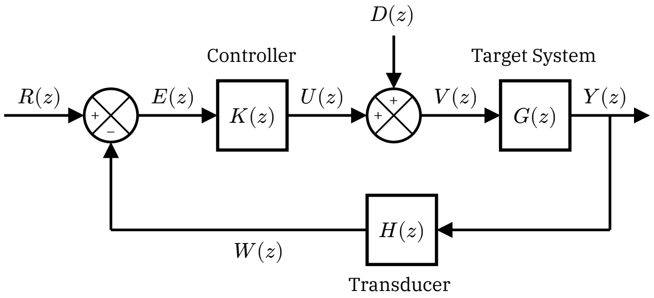


Figure 2. *Closed-loop control system with controller, plant, disturbance input, and feedback transducer.*

The disturbance $D(z)$ enters before the plant, while $H(z)$ shapes the measured feedback signal $W(z)$ returned to the summing junction.

Linear Algebra Diagram and Transition Graph

For a homomorphism $\varphi : G \to G'$, the **kernel** determines the quotient $G/\ker\varphi$, while the **image** is the subgroup of reachable outputs in G'.

$$\begin{tikzcd}
G \arrow[r, "\varphi"] \arrow[d, "-"'] & G' \\
G/\ker\varphi \arrow[r, "\overline{\varphi}"'] & \operatorname{im}\varphi \arrow[u, "\mathrm{inc}"']
\end{tikzcd}$$

Figure 3. *Kernel-image decomposition for a homomorphism $\varphi : G \to G'$.*

The induced map $\overline{\varphi}$ sends cosets modulo $\ker\varphi$ onto $\operatorname{im}\varphi$, and the inclusion embeds that image back into G'.

A weighted directed graph encodes reachable states as nodes and transition costs as labels on the arcs. This version keeps the notation compact to match the reference diagram.

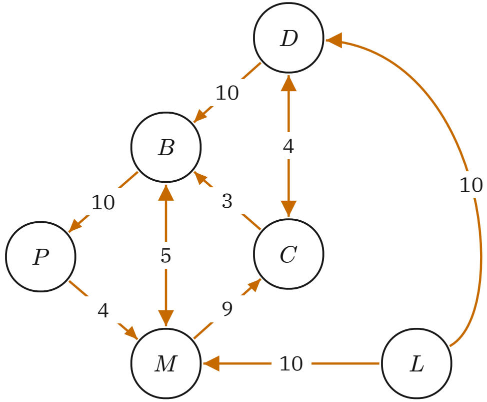
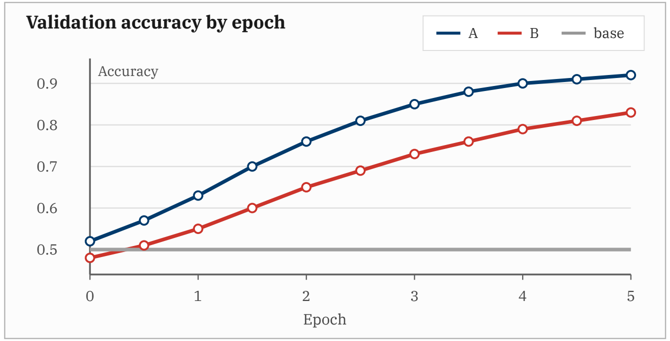


Figure 4. *Weighted directed transition graph; edge labels denote transition costs.*

Parallel and long-range transitions are shown with separate arrows, making bidirectional moves and high-cost paths visible at a glance.

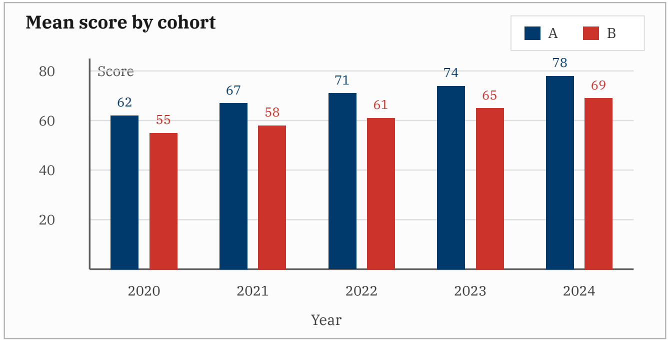
Model Accuracy and Cohort Scores (No Captions)

The accuracy curves compare Model A, Model B, and a 50% baseline across epochs $x \in [0, 5]$. Model A stays ahead throughout, while both learned models rise well above the baseline.



At epoch 5, **Model A** reaches 92% and **Model B** reaches 83%. Their gap grows up to epoch 3 and then narrows from 12 to 9 percentage points by the final epoch.

The grouped bars compare mean test scores for Group A and Group B from 2020 to 2024. Both cohorts improve each year, with **Group A** leading every annual pair.



Scores rise from 62 to 78 for **Group A** and from 55 to 69 for **Group B**. The gap widens from 7 points in 2020 to 9 points in 2024.

Model Accuracy and Cohort Scores (With Captions)

The accuracy curves compare Model A, Model B, and a 50% baseline across epochs $x \in [0, 5]$. Model A stays ahead throughout, while both learned models rise well above the baseline.

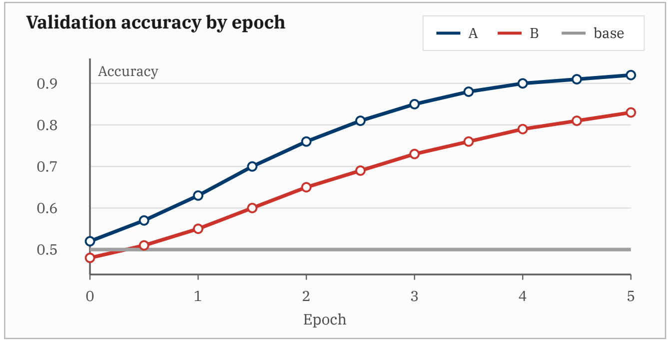


Figure 5. *Accuracy vs. epoch for Model A, Model B, and random baseline.*

At epoch 5, **Model A** reaches 92% and **Model B** reaches 83%. Their gap grows up to epoch 3 and then narrows from 12 to 9 percentage points by the final epoch.

The grouped bars compare mean test scores for Group A and Group B from 2020 to 2024. Both cohorts improve each year, with **Group A** leading every annual pair.

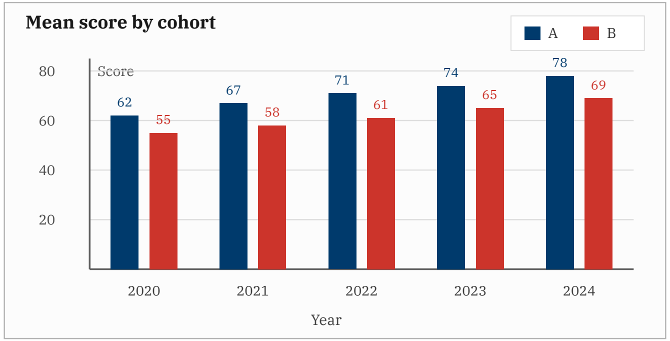


Figure 6. *Mean test score by group and year (2020-2024).*

Scores rise from 62 to 78 for **Group A** and from 55 to 69 for **Group B**. The gap widens from 7 points in 2020 to 9 points in 2024.

Section 8

Theorem-style Boxes

Definitions, Theorems, and Lemmas

Definition 8.1 Metric Space

A **metric space** (M, d) is a set M with $d : M \times M \to \mathbb{R}_{\geq 0}$ satisfying non-negativity, identity ($d(x, y) = 0 \Leftrightarrow x = y$), symmetry, and the triangle inequality.

Theorem 8.1 Banach Fixed-Point Theorem

Let (M, d) be complete and $f : M \to M$ a contraction with constant $k < 1$. Then f has a **unique** fixed point $x^* \in M$.

Lemma 8.1

Any contraction f on (M, d) is uniformly continuous and extends uniquely to the completion $\overline{M}$.

Corollary 8.1 Picard-Lindelof

Under Lipschitz continuity in y, the IVP $\dot{y} = f(t, y)$, $y(t_0) = y_0$ has a unique local solution.

Proof. Apply Banach's theorem to the Picard operator $T\varphi = y_0 + \int_{t_0}^{t} f(s, \varphi(s))ds$, which is contractive on $C([t_0 - \delta, t_0 + \delta])$ for small $\delta > 0$. ∎

Remark 8.1 Completeness is necessary

On $(0, 1)$ with $f(x) = \frac{x}{2}$, the fixed point 0 lies outside the space – Banach's theorem fails.

Examples, Exercises, Propositions, and Custom Boxes

Example 8.1 Euclidean Space

$\mathbb{R}^n$ with $d(x, y) = \|x - y\|_2$ is complete. The iteration $x_{k+1} = Ax_k + b$ converges iff $\rho(A) < 1$, to the unique solution of $x = Ax + b$.

Exercise 8.1

Show that $\mathbb{Z}_p$ is complete under the p-adic metric and use Banach's theorem to prove Hensel's lemma.

Proposition 8.1 Closed Subsets are Complete

A closed subset of a complete metric space is itself a complete metric space.

Note 8.2 Implementation tip

Monitor $\left\|x_{k+1} - x_k\right\|$ as a stopping criterion. A-priori error: $\|x_k - x^*\| \leq \frac{k^m}{1-k}\|x_1 - x_0\|$.

Warning 8.3 Common pitfall

Contraction ($k < 1$) is strictly stronger than nonexpansive ($k = 1$). Rotations on S^1 are nonexpansive but have no fixed points.

Custom 8.4 Any label works

Use `#fs-box("kind", name: "...", color: ...)` for any label and color – observations, facts, algorithms, warnings.

Custom Box Widths

Custom 8.5 Default width

This box uses the default width, so it fills the available slide content area. It is the recommended form when the material belongs to the main flow of the slide.

Custom 8.6 Narrow custom width

This box has a smaller explicit width. It is useful for short claims, reminders, or side notes that should not dominate the slide.

Custom 8.7 Short content, default width

A short phrase.

Full-Width Mathematical Proof

Theorem 8.8 Cauchy-Schwarz Inequality

For all vectors $u, v \in \mathbb{R}^n$,

$$|u \cdot v| \leq \|u\| \|v\|.$$

Proof. If $v = 0$, the claim is immediate, so assume $v \neq 0$. For every real t, the squared norm of $u - tv$ is non-negative:

$$0 \leq \|u - tv\|^2 = \|u\|^2 - 2t(u \cdot v) + t^2 \|v\|^2.$$

Choose the minimising value $t = \frac{u \cdot v}{\|v\|^2}$. Substitution gives

$$0 \leq \|u\|^2 - \frac{(u \cdot v)^2}{\|v\|^2}.$$

Multiplying by the positive number $\|v\|^2$, we obtain

$$(u \cdot v)^2 \leq \|u\|^2 \|v\|^2.$$

Taking square roots on both sides yields $|u \cdot v| \leq \|u\| \|v\|$. Equality occurs precisely when the non-negative quadratic has a zero at its minimum, which means $u - tv = 0$ for some scalar t; equivalently, the two vectors are linearly dependent. ∎

Narrow Mathematical Proof

Proof environments fill the available content width by default, matching the rhythm of ordinary slide material. When a shorter line length reads better, pass an explicit `width` and center the proof as below.

Proof. We prove Young's inequality in its weighted quadratic form. Let $a, b \geq 0$ and fix $\varepsilon > 0$. Since every square is non-negative,

$$0 \leq \left(\sqrt{\varepsilon}a - \frac{b}{\sqrt{\varepsilon}}\right)^2.$$

Expanding the square gives

$$0 \leq \varepsilon a^2 - 2ab + \frac{b^2}{\varepsilon}.$$

Rearranging terms and dividing by 2 yields

$$ab \leq \varepsilon \frac{a^2}{2} + \frac{b^2}{2\varepsilon}.$$

This form is especially useful when a product term must be absorbed into a coercive estimate: one chooses ε small enough for the first term and pays for it in the second term. ∎

Theorem Box with Internal Proof

Theorem 8.9 Cauchy-Schwarz Inequality

For all vectors $u, v \in \mathbb{R}^n$,

$$|u \cdot v| \leq \|u\|\|v\|.$$

Proof. If $v = 0$, the claim is immediate. Otherwise, the quadratic expression $\|u - tv\|^2$ is non-negative for every $t \in \mathbb{R}$. Expanding and choosing $t = \frac{u \cdot v}{\|v\|^2}$ gives

$$0 \leq \|u\|^2 - \frac{(u \cdot v)^2}{\|v\|^2}.$$

Multiplying by $\|v\|^2$ and taking square roots yields the desired inequality. ■

Exercise Box with Solution

Exercise 8.2 Spectral radius criterion

Let $A \in \mathbb{R}^{n\times n}$ and suppose $\|A\| < 1$ for a matrix norm compatible with the vector norm. Prove that $I - A$ is invertible and derive a convergent series for its inverse.

Solution

Since $\left\|A^k\right\| \leq \|A\|^k$, the Neumann series $\sum_{k=0}^{\infty} A^k$ converges absolutely. Multiplying partial sums by $I - A$ gives $I - A^{m+1}$, which tends to I. Thus,

$$(I - A)^{-1} = \sum_{k=0}^{\infty} A^k.$$

Exercise Box with Two Solutions

Exercise 8.3 Arithmetic-geometric mean

Prove that for $x, y \geq 0$ one has $\sqrt{xy} \leq \frac{x+y}{2}$.

Solution 1

The square $(\sqrt{x} - \sqrt{y})^2$ is non-negative, so $x + y - 2\sqrt{xy} \geq 0$.

Solution 2

The function $\log$ is concave on $(0, \infty)$. Applying Jensen's inequality to x and y gives

$$\log\left(\frac{x+y}{2}\right) \geq \frac{\log x + \log y}{2} = \log(\sqrt{xy}).$$

Theorem, Text, and Lemma

The next result is stated in the same theorem-style box used throughout the template. The surrounding prose is deliberately included to show how normal paragraphs sit before, between, and after formal statements.

Theorem 8.10 Compactness criterion

Every sequence in a compact metric space has a convergent subsequence whose limit belongs to the same space.

This theorem is often the bridge between qualitative assumptions and quantitative estimates. The intermediate text lets the slide show how ordinary prose separates theorem-style boxes without requiring a two-column layout.

Lemma 8.2 Closed image of a convergent sequence

If $x_n \to x$ and F is closed, then every sequence contained in F can only converge to a point of F.

Together, the theorem and lemma give a compact workflow: first extract convergence, then use closedness to keep the limiting object inside the admissible set. This final paragraph checks the lower spacing after the last formal box.

Thank you for your attention